Anti-Sleep and Alcohol Detection System

ESP32 DevKit • Arduino IDE • MQ-3 Alcohol Sensor • Eye Blink Sensor • L298N Motor Driver

Complete Arduino Sketch

The code below reads the MQ-3 analog value and eye-blink sensor output, activates the buzzer and stops the motor when an alert condition is detected.

```
/*
  Project: Anti-Sleep and Alcohol Detection System
  Controller: ESP32 DevKit
  Sensors: MQ-3 Alcohol Sensor, Eye Blink Sensor
  Driver: L298N
*/

const int MQ3_PIN = 34;
const int EYE_PIN = 27;
const int BUZZER_PIN = 14;

// L298N motor driver pins
const int ENA = 33;
const int IN1 = 26;
const int IN2 = 25;

// Adjust after checking sensor readings in Serial Monitor
int alcoholThreshold = 1800;

// Change to HIGH if your sensor outputs HIGH when eyes are closed
const int EYES_CLOSED_LEVEL = LOW;

// Eyes must remain closed this long to trigger an alarm
const unsigned long eyeClosedLimit = 1500;

unsigned long eyeClosedStart = 0;
bool eyeTimerStarted = false;

void stopMotor() {
  digitalWrite(ENA, LOW);
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
}

void runMotor() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(ENA, HIGH);
}

void setup() {
  Serial.begin(115200);

  pinMode(EYE_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  digitalWrite(BUZZER_PIN, LOW);
  stopMotor();

  Serial.println("Anti-Sleep and Alcohol Detection System");
  Serial.println("Warming up MQ-3 sensor...");

  // Allow the sensor to stabilize; initial readings may be unreliable
  delay(20000);

  Serial.println("System ready.");
}

void loop() {
  int alcoholValue = analogRead(MQ3_PIN);
  int eyeValue = digitalRead(EYE_PIN);

  bool alcoholDetected = alcoholValue >= alcoholThreshold;
  bool eyesClosed = eyeValue == EYES_CLOSED_LEVEL;

  // Detect sustained eye closure
  if (eyesClosed) {
    if (!eyeTimerStarted) {
      eyeClosedStart = millis();
      eyeTimerStarted = true;
    }
  } else {
    eyeTimerStarted = false;
```

```
  }

  bool sleepyDetected =
      eyeTimerStarted &&
      (millis() - eyeClosedStart >= eyeClosedLimit);

  // Display sensor readings for calibration
  Serial.print("MQ-3: ");
  Serial.print(alcoholValue);
  Serial.print(" | Eye sensor: ");
  Serial.print(eyeValue);

  if (alcoholDetected || sleepyDetected) {
    stopMotor();
    digitalWrite(BUZZER_PIN, HIGH);

    if (alcoholDetected && sleepyDetected) {
      Serial.println(" | ALCOHOL + SLEEP ALERT!");
    } else if (alcoholDetected) {
      Serial.println(" | ALCOHOL DETECTED!");
    } else {
      Serial.println(" | EYES CLOSED - SLEEP ALERT!");
    }
  } else {
    runMotor();
    digitalWrite(BUZZER_PIN, LOW);
    Serial.println(" | Normal - Motor ON");
  }

  delay(100);
}
```

Important setup notes

- Set the Arduino IDE board and COM port for your ESP32 before uploading.
- Open Serial Monitor at 115200 baud. Adjust `alcoholThreshold` using observed sensor readings; 1800 is only an example.
- Change `EYES_CLOSED_LEVEL` to HIGH if your eye sensor outputs HIGH when eyes are closed.
- MQ-3 analog output must be kept at or below 3.3 V at GPIO34; use a suitable voltage divider.
- All grounds must be common. Power the motor through the L298N, not from an ESP32 GPIO.
- A 20-second delay is only a preliminary warm-up; the MQ-3 may need several minutes to stabilize.
- This is a prototype demonstration, not a certified vehicle safety system. Test with the motor unloaded and secured.
- Connect a buzzer directly to GPIO14 only if its voltage/current ratings are safe for the ESP32 pin; otherwise use a suitable driver.